EE2016 Microprocessor Theory and Lab

EE Dept – *IIT Madras Fall, 2025*

Dr. R. Manivasakan

Presentation: Week 1 & 2

Introduction

EE2016 Dr. R. Manivasakan, OSWM lab, EE Dept, IITM:
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Course Information

- Course Information
  - Credit Scheme
  - Syllabus (Curricula)
  - Text Books
    - William Stallings, “Computer Architecture and Organization: 8th Edition”,
    - Mazidi, “The AVR Microcontroller and Embedded System, using Assembly and C”
    - ARM Assembly ……
    - Welsh, “ARM Assembly Language Programming”
  - References
    - David A. Patterson and John L. Hennessy, “Computer Organization and Design: ...”, Elsevier, Third Edition, 2005
    - Carl Hamacher, Zvonko Vranesic and Safwat Zaky, “Computer Organization”, Tata McGraw Hill, Fifth Edition, 2002
    - J P Hayes, “Computer Architecture and Organization”, 3ed, McGrawHill
  - Lab: (1 FPGA), 4 AVR Microcontrollers, 3 ARM based experiments

Course Information

- Syllabus

Mainly microcontroller, microprocessor concepts, design, architecture: CU, EU, buses, I/O, memory, Van Neumann / Harvard architecture, performance assessment, memory organization, cache management. Lab: Hands-on-assembly programming: AVR (8-bit) microcontroller and ARM 32-bit processor

- MuP bottom most stack in engineering – implementation of any engineering idea, theory --> practice to the core
- Positioning of EE2016
  - Physics--> Electronics-->Digital Circuits--> Digital Systems--> EE2016
  - Future Courses
    - Computer Organization
    - VLSI Design --> Algorithms for VLSI Design
    - VLSI Technology
    - Hardware based courses in EE?

Course Information

- Computer Organization Vs Microprocessor Theory & Lab
- EE2003 Computer organization
  - More of a system level – software, compiler, OS, I/O
- Microprocessor
  - More of microprocessor / microcontroller / embedded system level study
  - May be complete working system (microcontroller / embedded system) but scaled down version
  - CU & EU – centric and Assembly programming rather than C (embedded C programming).

Credit Scheme (Tentative)

- Credit scheme
  - 50 % theory 50 % lab
- Theory (50 %)
  - 25 % - Q1 (12) + Q2 (13)
  - 25 % End semester theory
- Lab (50 %)
  - Regular Lab Session (30 %)
  - Endsem Lab Exam (20 %)

Lecture 1: Overview

- Microprocessors, Microcontrollers & other closely related terms
- Applications of Microprocessor / Microcontrollers
  - Mother of ALL engineering disciplines
    - Wherever implementation of an engineering concept comes, muP / muC would be there
- Microprocessors
  - Definition
  - Architecture and Organization
  - Applications
- Categorization of Microprocessor / Computer Systems
- RISC or CISC
- Van-Neumann architecture and Harvard architecture

Lecture 1

- Microcontroller technology is the mother of ALL disciplines of engineering as it implements theory
  - All theory developed, have to be implemented, in the real world.
  - Implementation through microcontrollers
- Microprocessors & Microcontrollers
  - Commercial muPs/ muControllers, companies
  - Difference between muPs and muControllers
  - Closely related terms: FPGA, ASIC, SoC, CoM, SoM, Embedded processor, Embedded systems, microcomputer, server, work station, main frame computer, super computer and HPCs etc
- Overview of Microprocessors
  - Architecture and Organization
  - Applications of Microprocessors
    - Microcontroller, microcomputer, server, work station, main frame & HPCs
  - Microcontroller Applications
    - Embedded systems, SoC, wearables, IoT etc

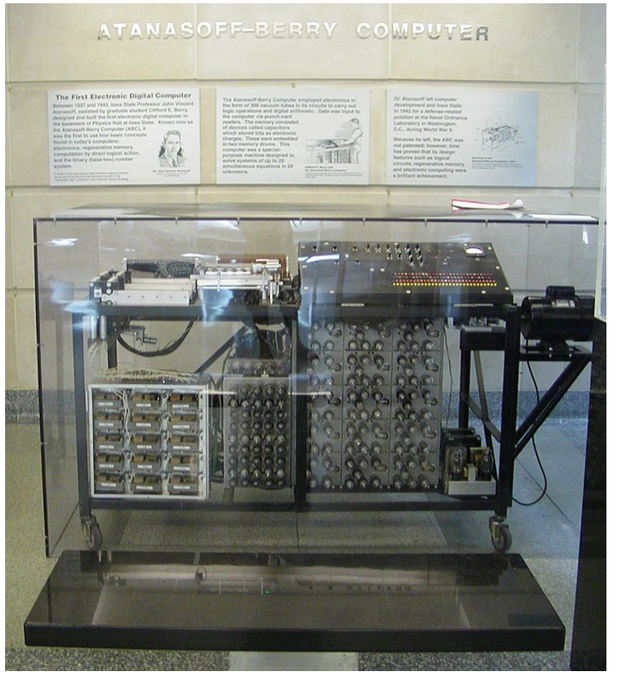
Microprocessor

- A Microprocessor
  - Is a multi-purpose clock driven, register based digital IC of logical and arithmetic computation capability, that accepts binary input, processes it according to instructions stored in memory (externally connected to it) and provides results as output (via I/O module externally connected to it).
    - Combines combinational and sequential logic
    - Operate on numerals and symbols represented in binary numeral systems

- Before microprocessors,
- Zeroth Generation
  - Electromechanical relays
    - Z3 – Konrad Zuse, Germany
- 1st Generation
  - Vaccum tubes
  - (1945) ENIAC – electronic, digital, programmable, general purpose
  - Vaccum tube (1951) IAS van Neumann first stored program
- 2nd Generation
  - Solid state but discrete transistor 1954 (TRADIC – Transistor Dgtl C)
- 3rd Generation
  - First IC based computer in 1964 UNIVAC 1108

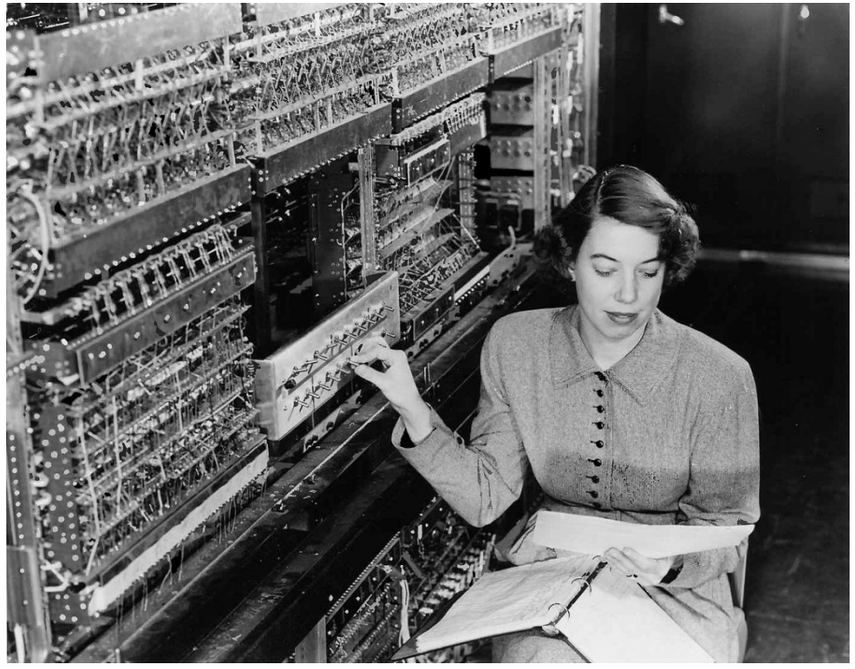
The 1946 ENIAC computer used more than 17,000 vacuum tubes



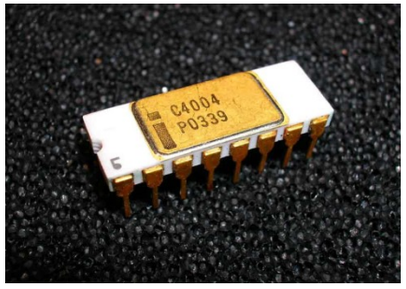


Replica of the Atanasoff–Berry computer at Iowa State University (1942)

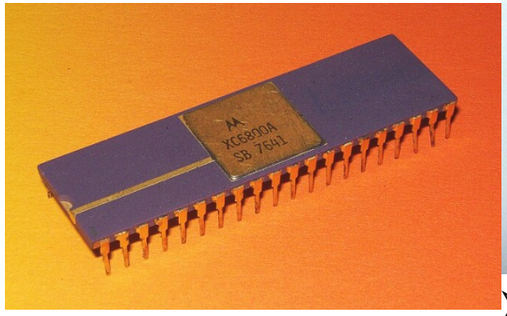
Legacy / Vintage Computers

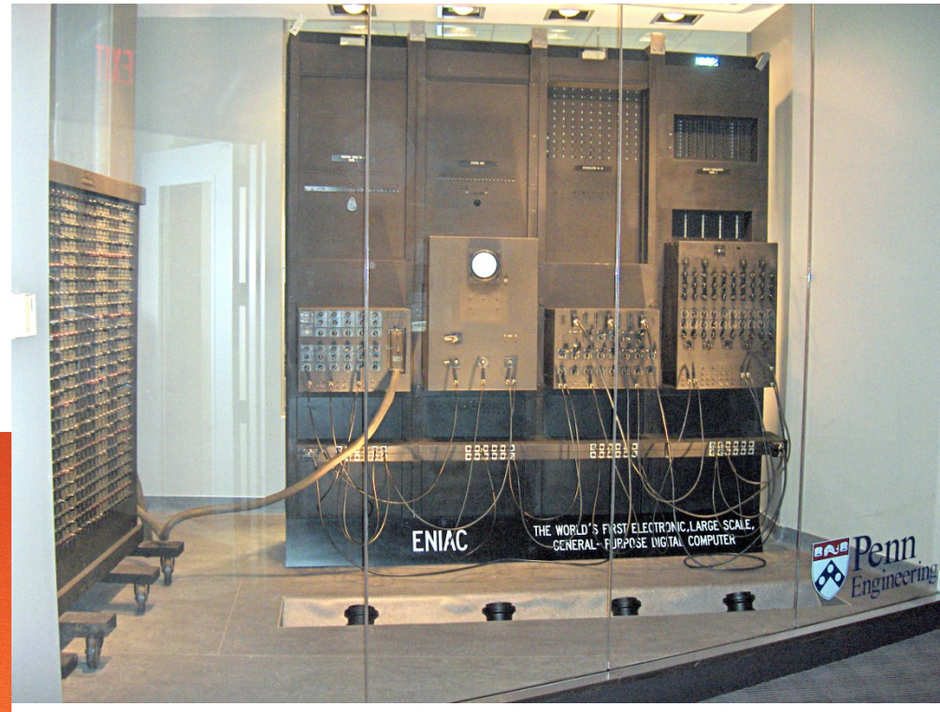


AVIDAC (1953) (vaccum tube) partially based on IAS machine developed by van Neumann

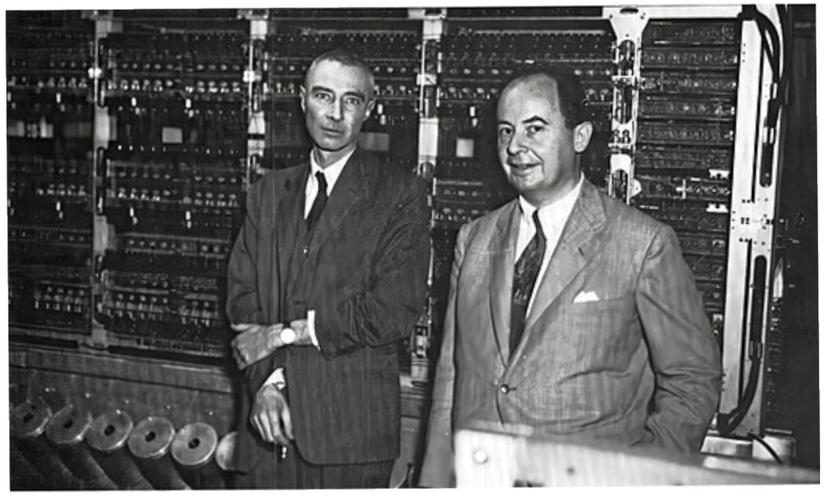


Intel's first processor 4004! (1971)

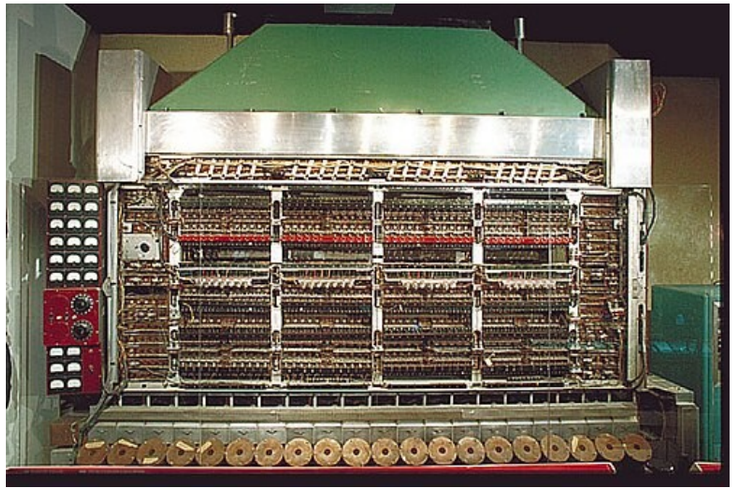




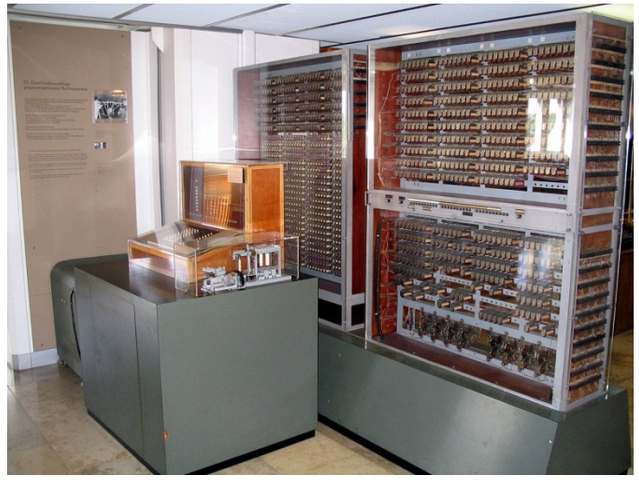
ENIAC computer (1945) vaccum tube based



Oppenheimer with van Neumann in front of IAS machine (1951)



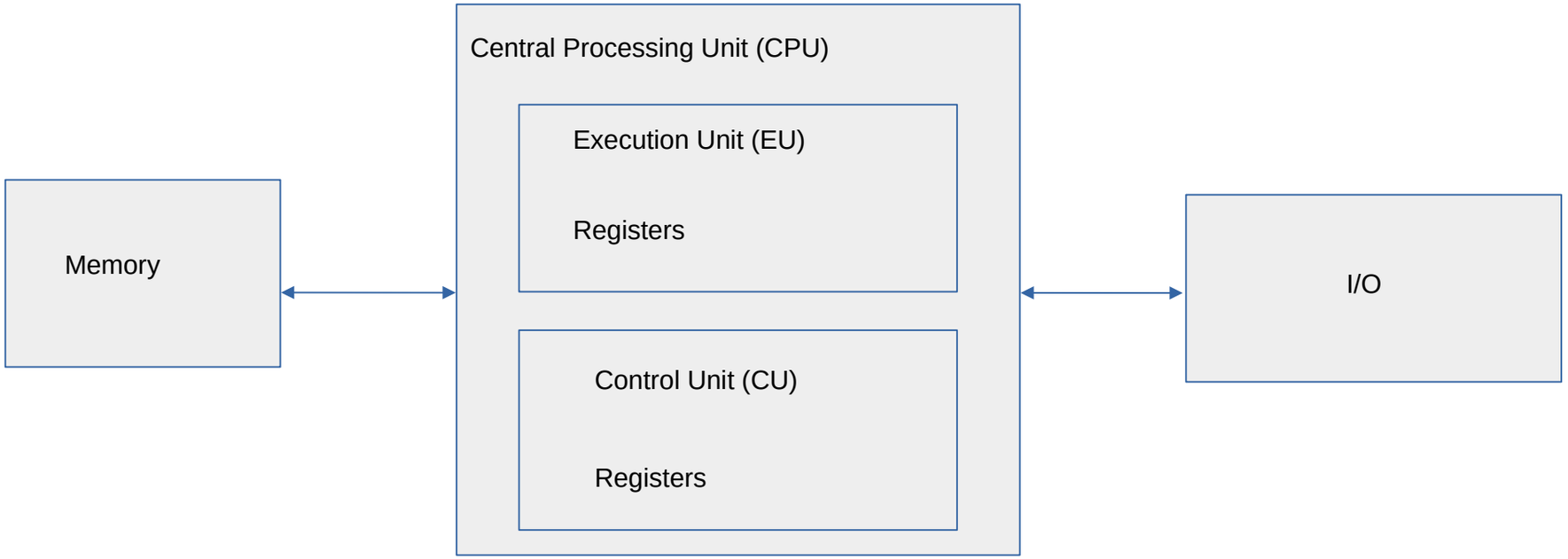
IAS machine (1951), Princeton University

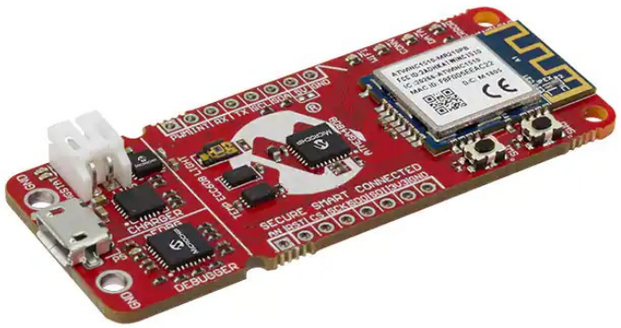


Electromechanical Z3 (1941) by Konrad Zuse, Germany

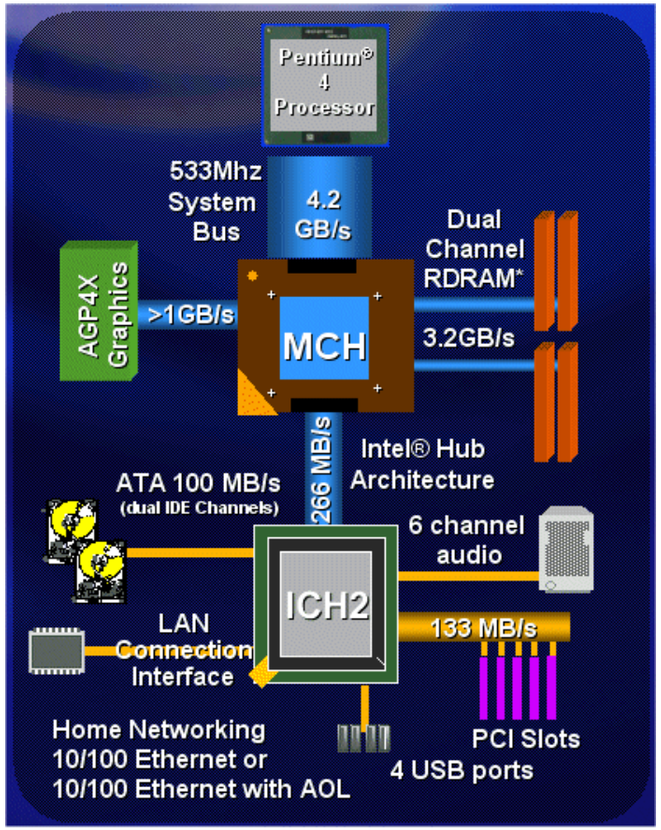
Microcomputer

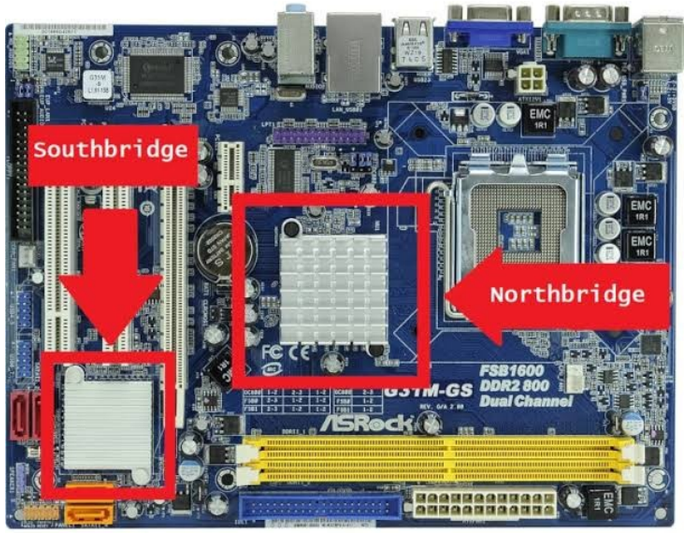
- A microcomputer is built over a microprocessor
- I/O and memory are interfaced to muP to get a working form - microcomputer





Ready to use sensor module
(AVR-IoT module)





A Desktop Motherboard: Block Diagram.



A Pluggable Computer on
Module (CoM)

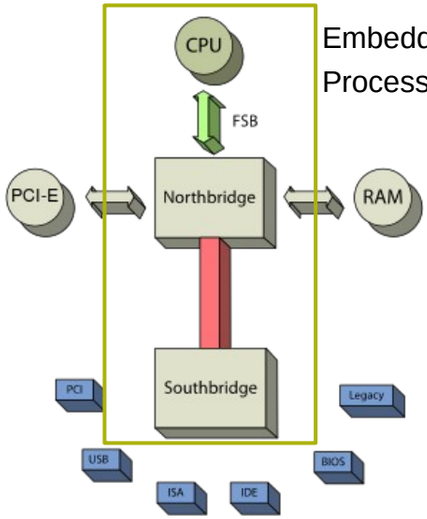
MCH – Memory Controller Hub
ICH2 – I/O Controller Hub



A desktop computer



Desktop motherboard



Embedded
Processor



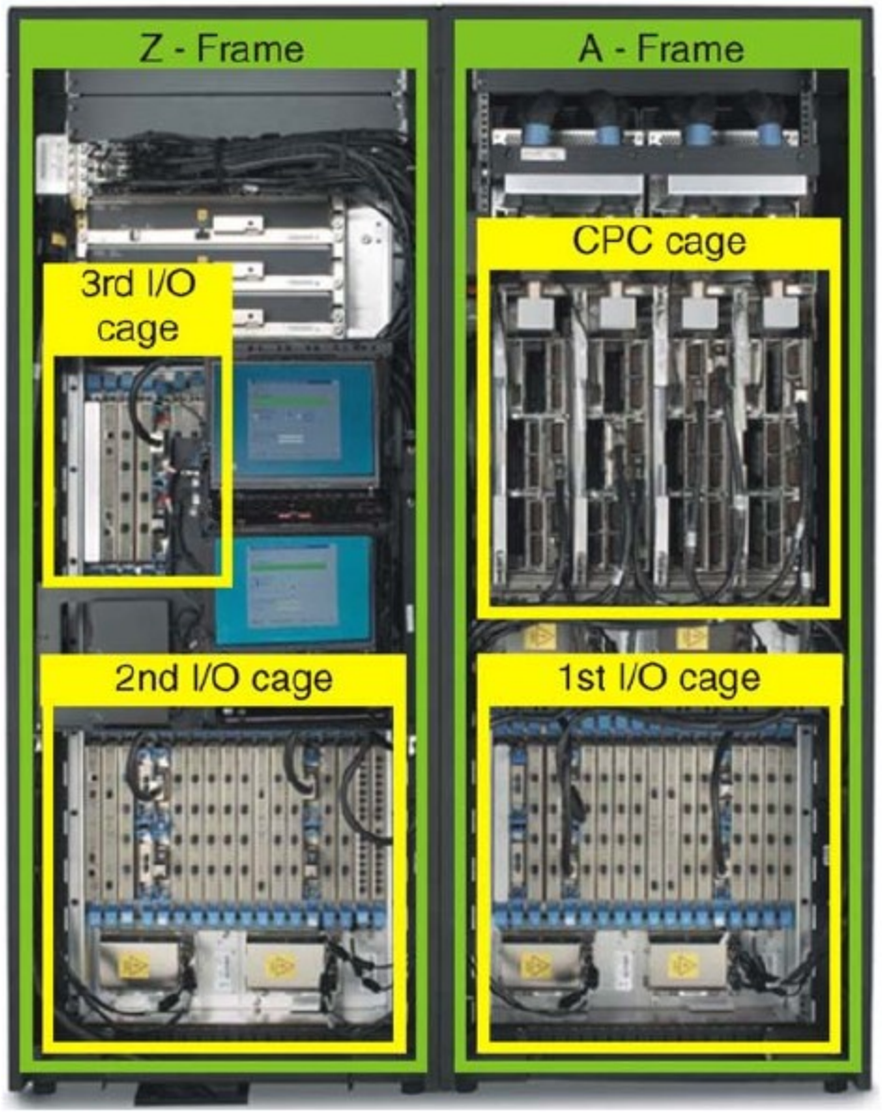
An embedded
Motherboard (only
CPU slot with I/O



Super computer



An IBM z mainframe computer

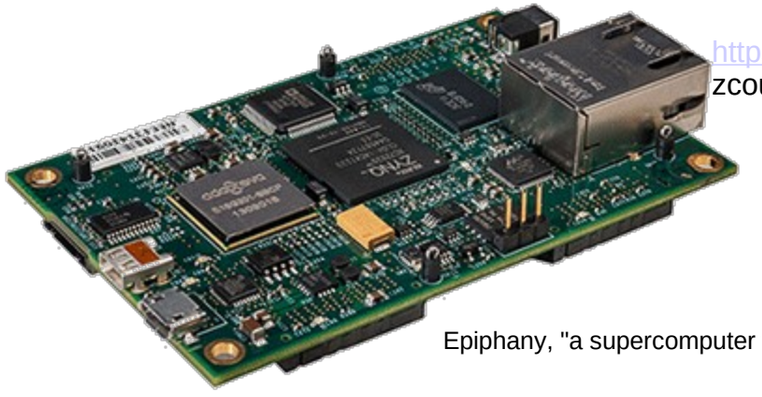


An IBM z mainframe
Computer: Internals

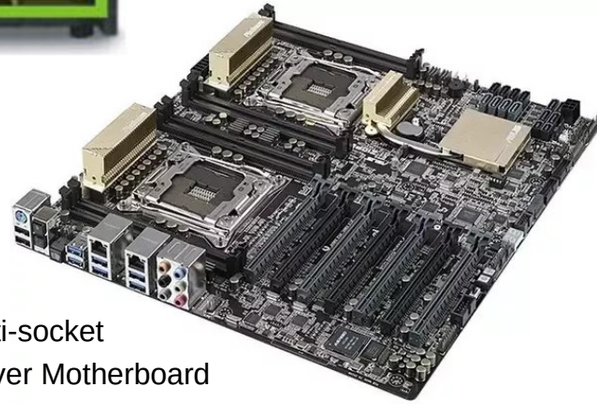


Workstation

https://www.ibm.com/docs/en/zosbasics/com.ibm.zos.zcourses/
zcourses_MFHWinternals.pdf

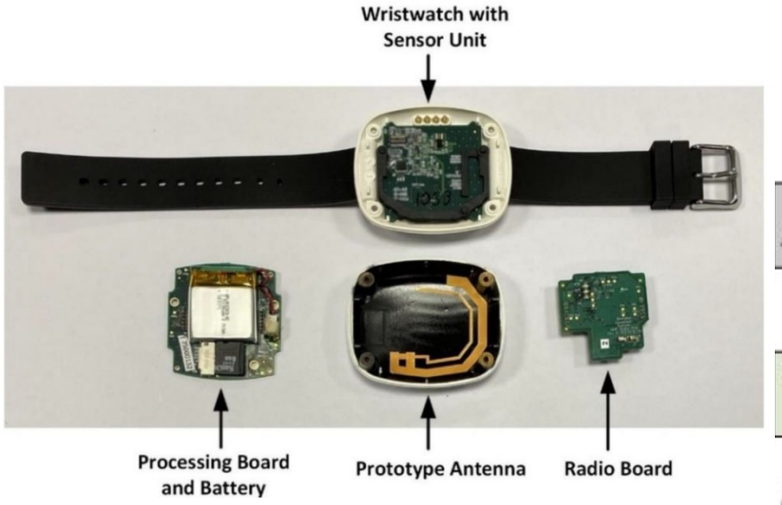


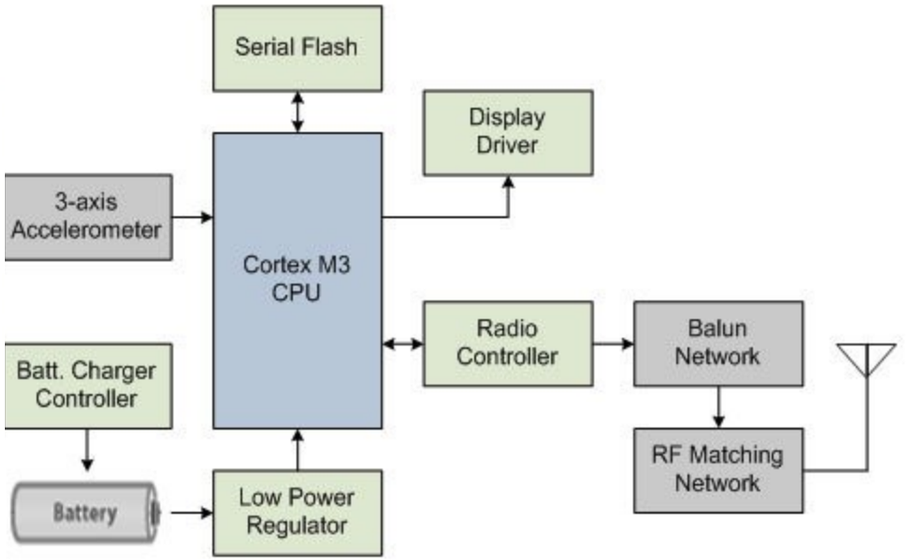
Epiphany, "a supercomputer for everyone"

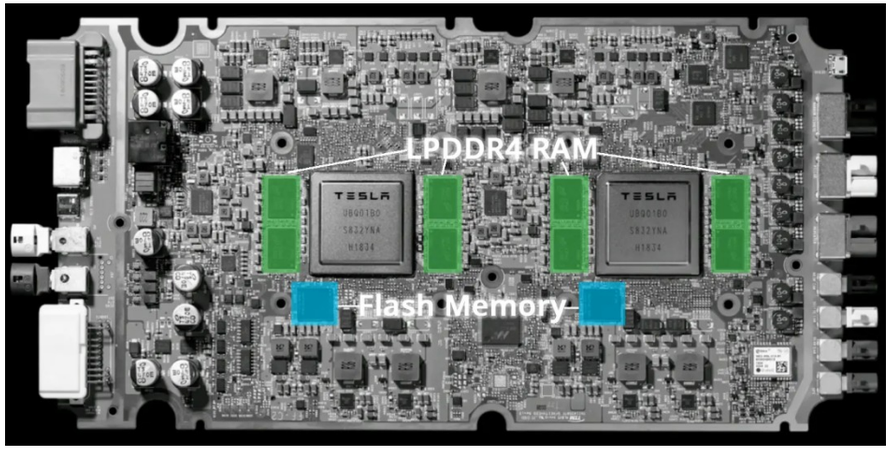


Multi-socket
Server Motherboard





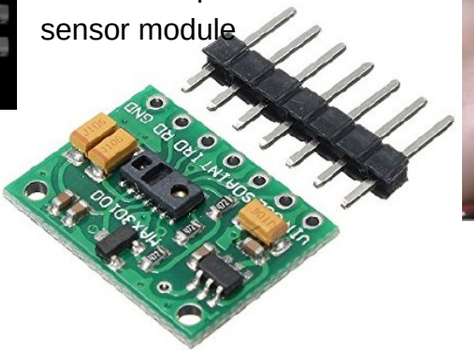




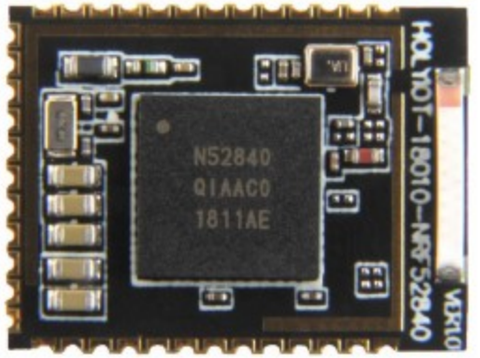
Teslas New HW3 Self-Driving Computer. Its A Beast



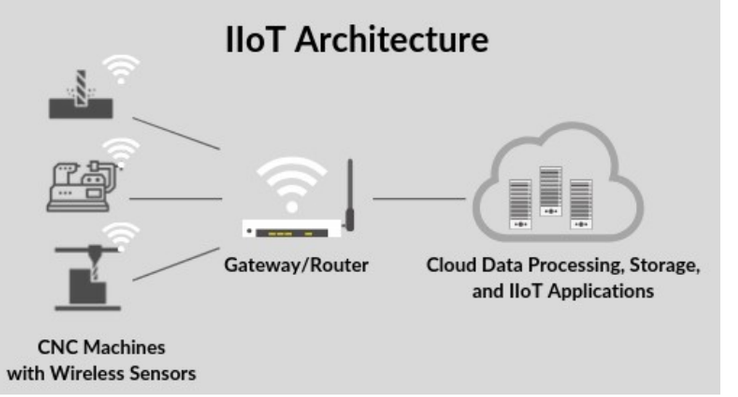
MAX30100 pulse oximeter sensor module







nRF52840 low power BLE module with ceramic antenna



Wearable --> IoT --> Microcontroller --> Micro-computer --> workstation --> server --> main frame computer --> super computer

Microprocessors & Microcontrollers

- Also called CPU

Intel 4004 (1971)



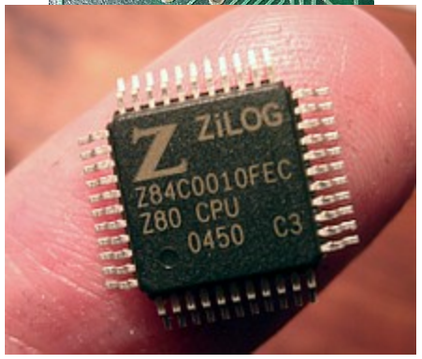


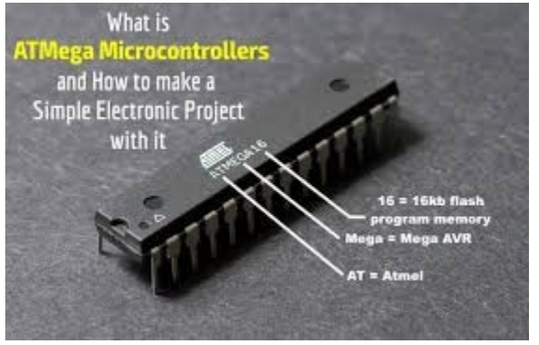
















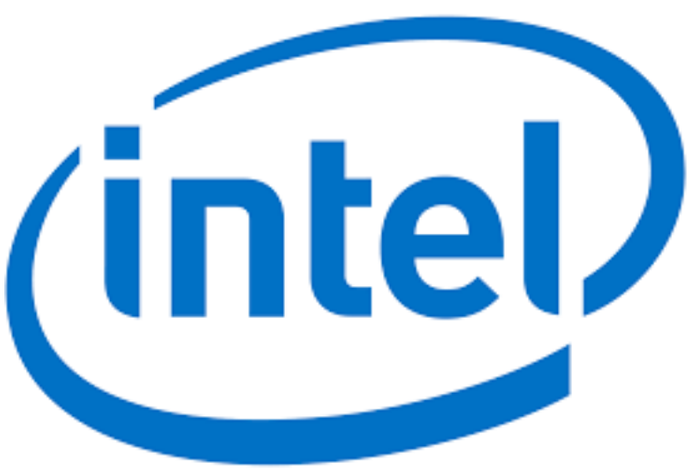








Companies Manufacturing Microprocessors































Companies Manufacturing Microcontrollers























Applications of Microcontrollers









































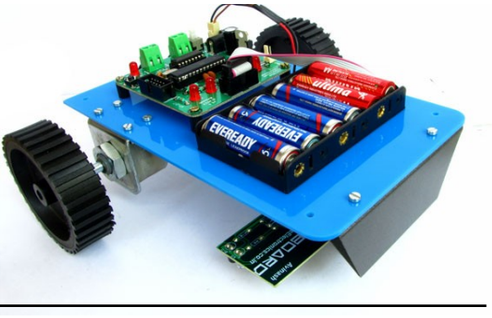






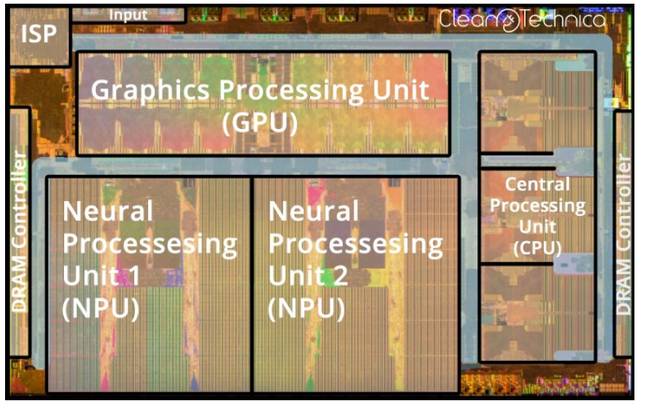






Complementary CPUs (Microprocessors)

- Central Processing Unit (CPU)
  - General purpose
- Specific Processing Units
  - Graphics Processing Unit
    - Processes graphics only
  - Neural Processing Unit (NPU)
    - Machine learning processor
  - Tensor Processing Unit (TPU)
  - Floating Point computation Unit (FPU)
  - Accelerated Processing Unit (APU) – AMD
    - CPU + GPU
  - Vision Processing Unit (VPU)
    - Same as GPU ~ similar
  - Parallel Processing Unit (PPU)
  - Digital Signal Processor (DSP)
    - Hardware engines specifically do the basic signal processing routines



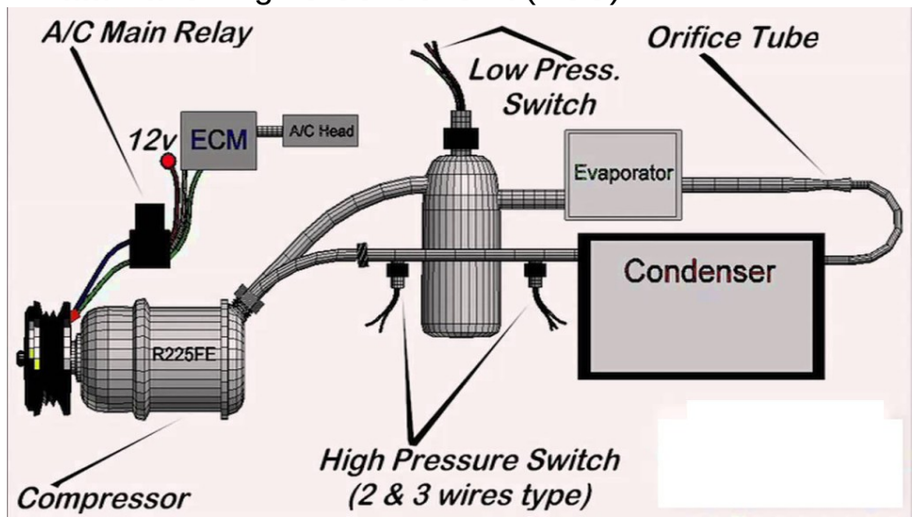
Applications of Microcontrollers

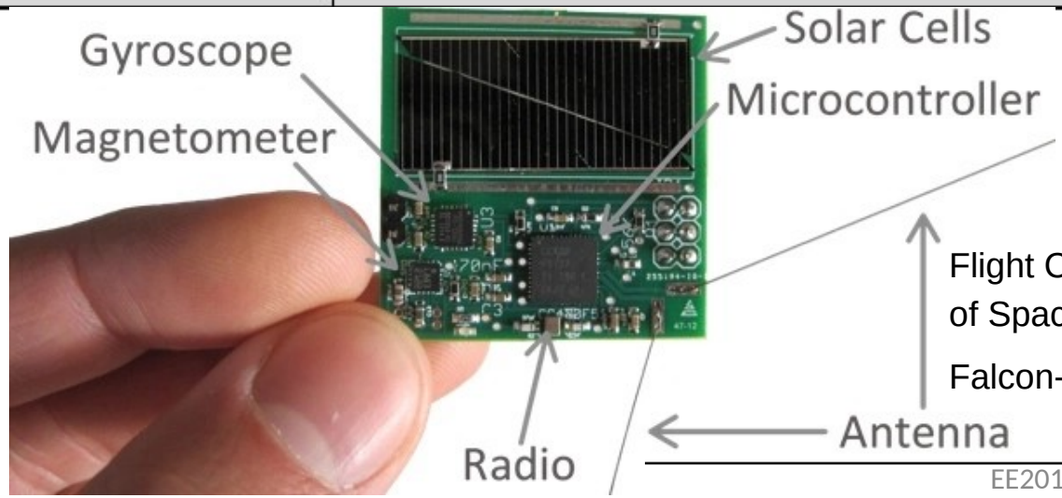
Market	Embedded Device
Automotive	Ignition system Engine control Brake system
Consumer electronics	Digital and analog televisions Set-top boxes (DVDs, VCRs, Cable boxes) Personal digital assistants (PDAs) Kitchen appliances (refrigerators, toasters, microwave ovens) Automobiles Toys/games Telephones/cell phones/pagers Cameras Global positioning systems
Industrial control	Robotics and controls systems for manufacturing Sensors
Medical	Infusion pumps Dialysis machines Prosthetic devices Cardiac monitors
Office automation	Fax machine Photocopier Printers Monitors Scanners





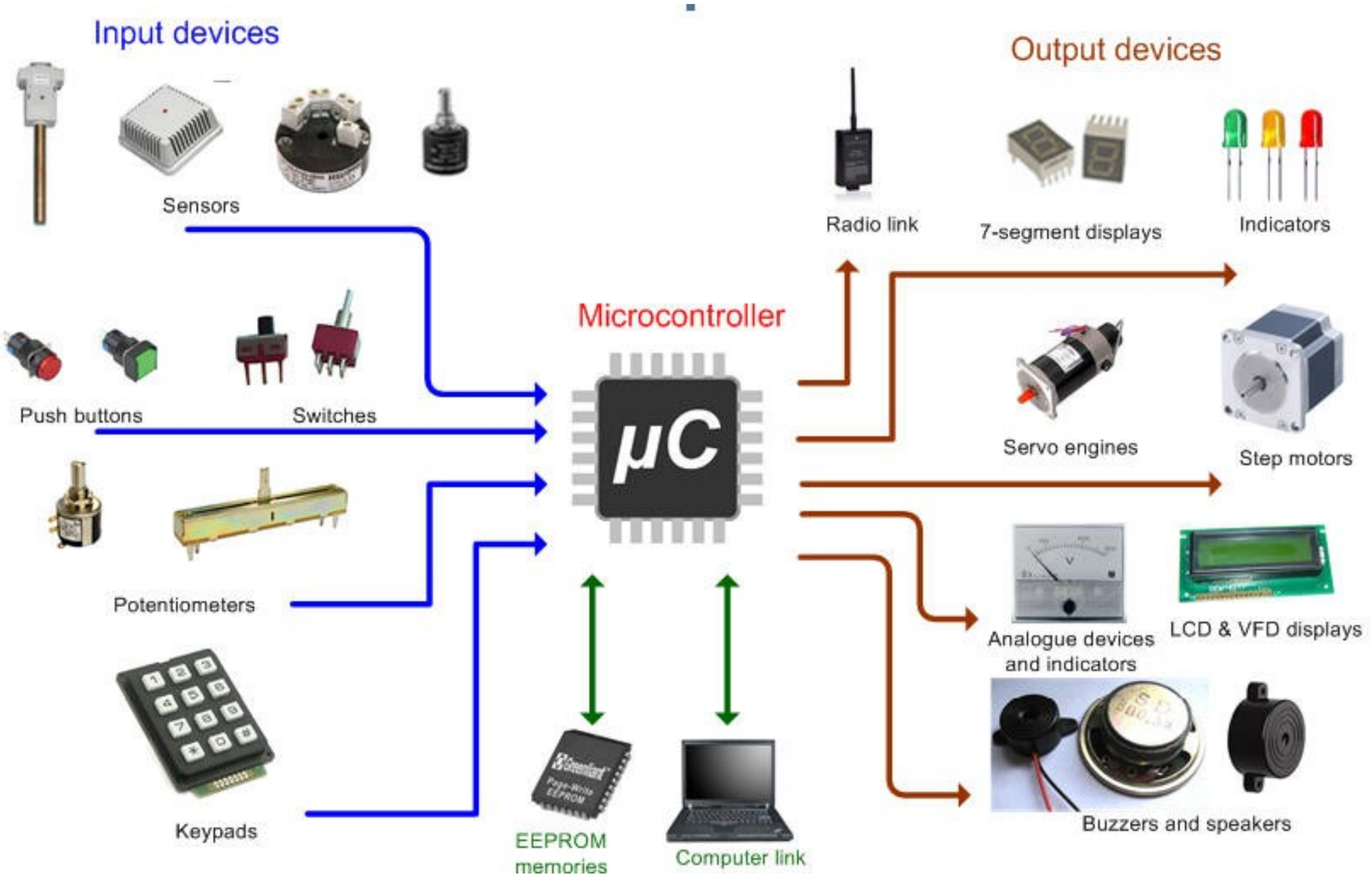
Tata Nano Engine Control Unit (ECU)





Xn+1 = Xn + rho_n




Input devices
Sensors
Push buttons
Switches
Potentiometers
Keypads
Microcontroller
µC
EEPROM memories
Computer link
Output devices
Radio link
7-segment displays
Indicators
Servo engines
Step motors
Analogue devices and indicators
LCD & VFD displays
Buzzers and speakers

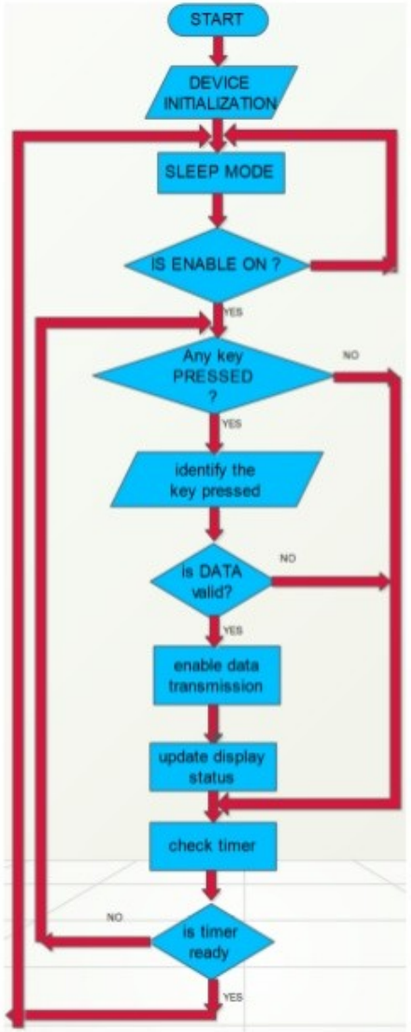
Applications of Microcontrollers: Process the Data from Sensor Modules & control

- Temperature sensor module
- Vibration switch module
- Hall magnetic sensor module
- Key switch module
- Infrared emission sensor module
- Laser sensor module
- Small passive buzzer module
- 3-color full-color LED SMD modules
- Photo interrupter module
- 2-color LED module
- Active buzzer module
- Temperature sensor module
- Temperature & humidity sensor module
- 3-color LED module
- Mercury open optical module
- Photo resistor module
- 5V relay module
- Finger measuring heartbeat module
- Tilt switch module
- Mini magnetic reed modules
- Infrared sensor receiver module
- XY-axis joystick module
- Linear magnetic Hall sensors
- Reed module
- Flame sensor module
- Magic light cup module
- Temperature sensor module
- 5mm red and green LED (common cathode) module
- Knock sensor module
- Obstacle avoidance sensor module
- Automatic flashing colorful LED module
- Analog Hall magnetic sensor module
- Metal touch sensor module
- Sensitive small microphone sensor module
- Sensitive Big microphone sensor module



Microcontroller Applications: Example

Remote control software program flowchart



Receiver side software program flowchart

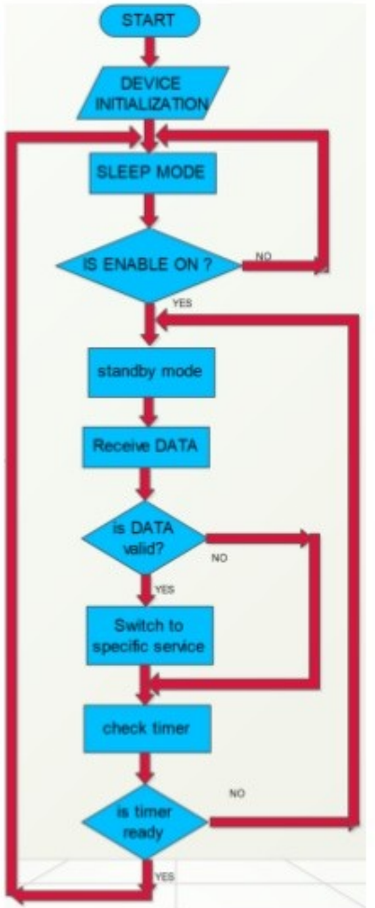
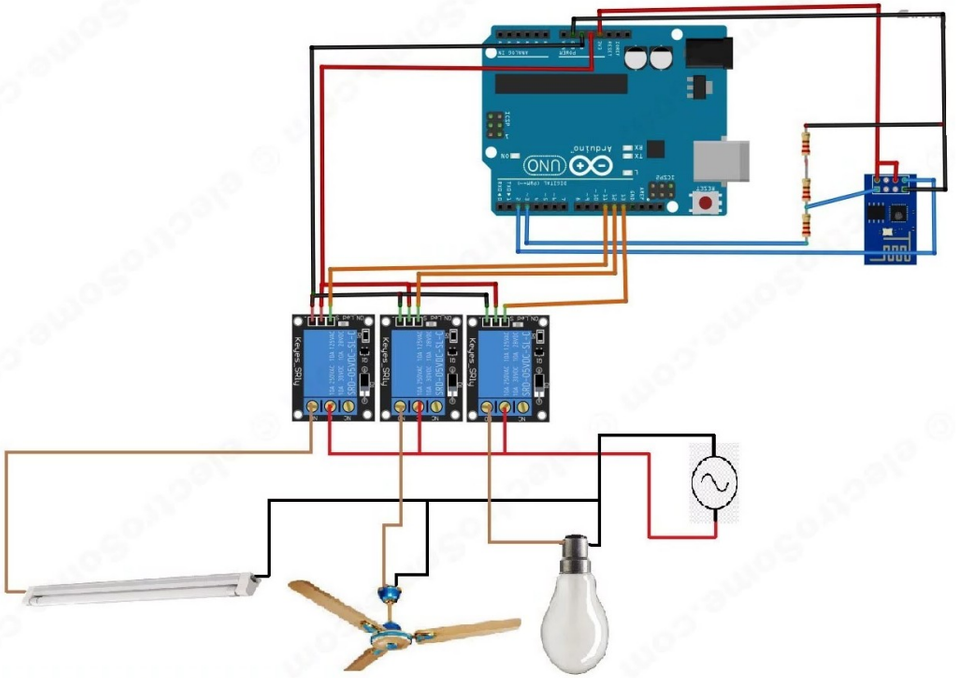
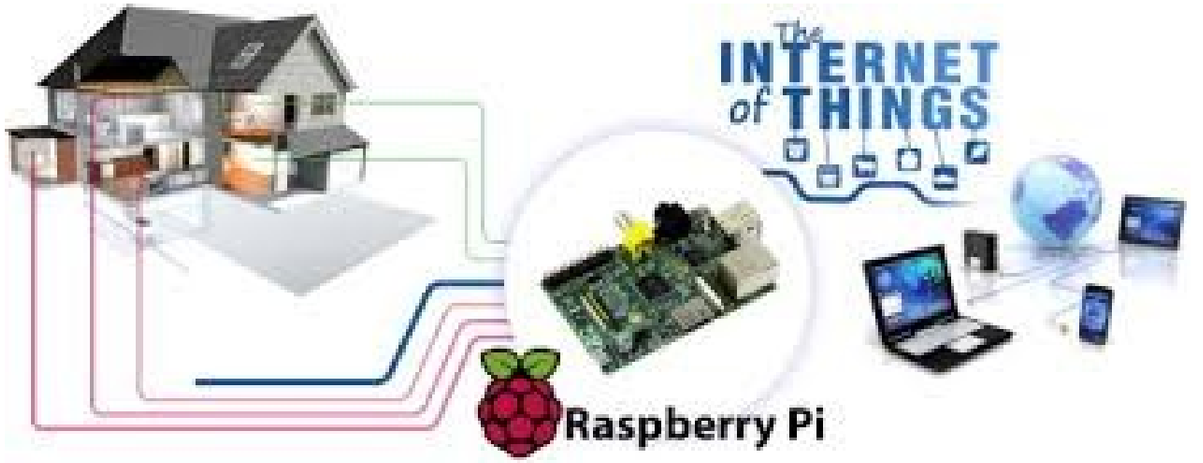


Figure 2 Flow charts for software programs for transmitter and receiver microcontrollers

WIRELESS HOME AUTOMATION USING PIC MICROCONTROLLER BASED ON RF-MODULE

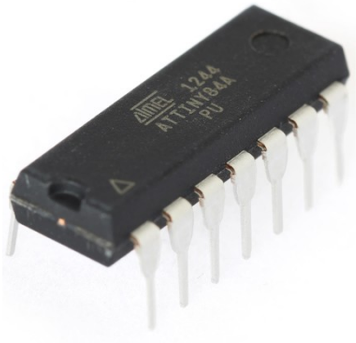




Atmel AVR Processors



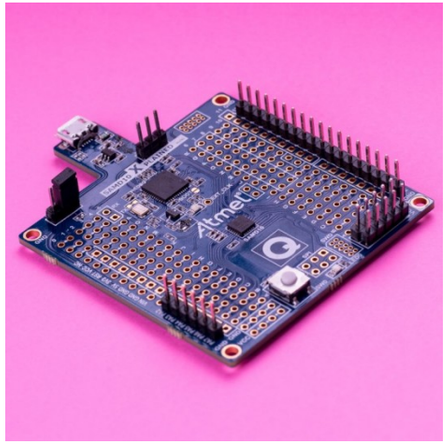




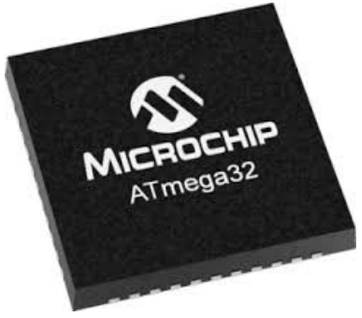


Amazing 1$ Microcontroller







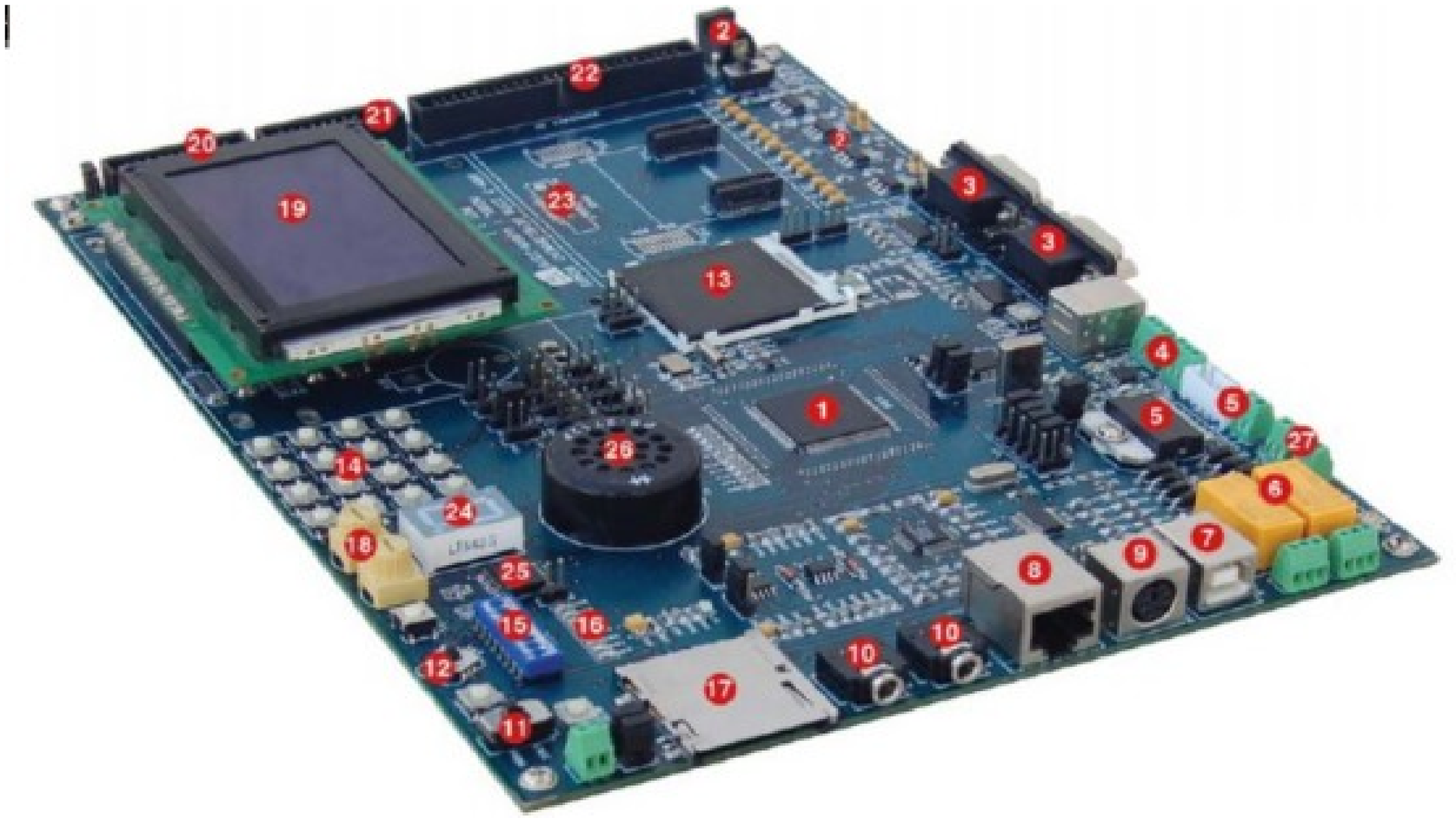




Vi Microsystem's ViARM LPC2378 Development Board



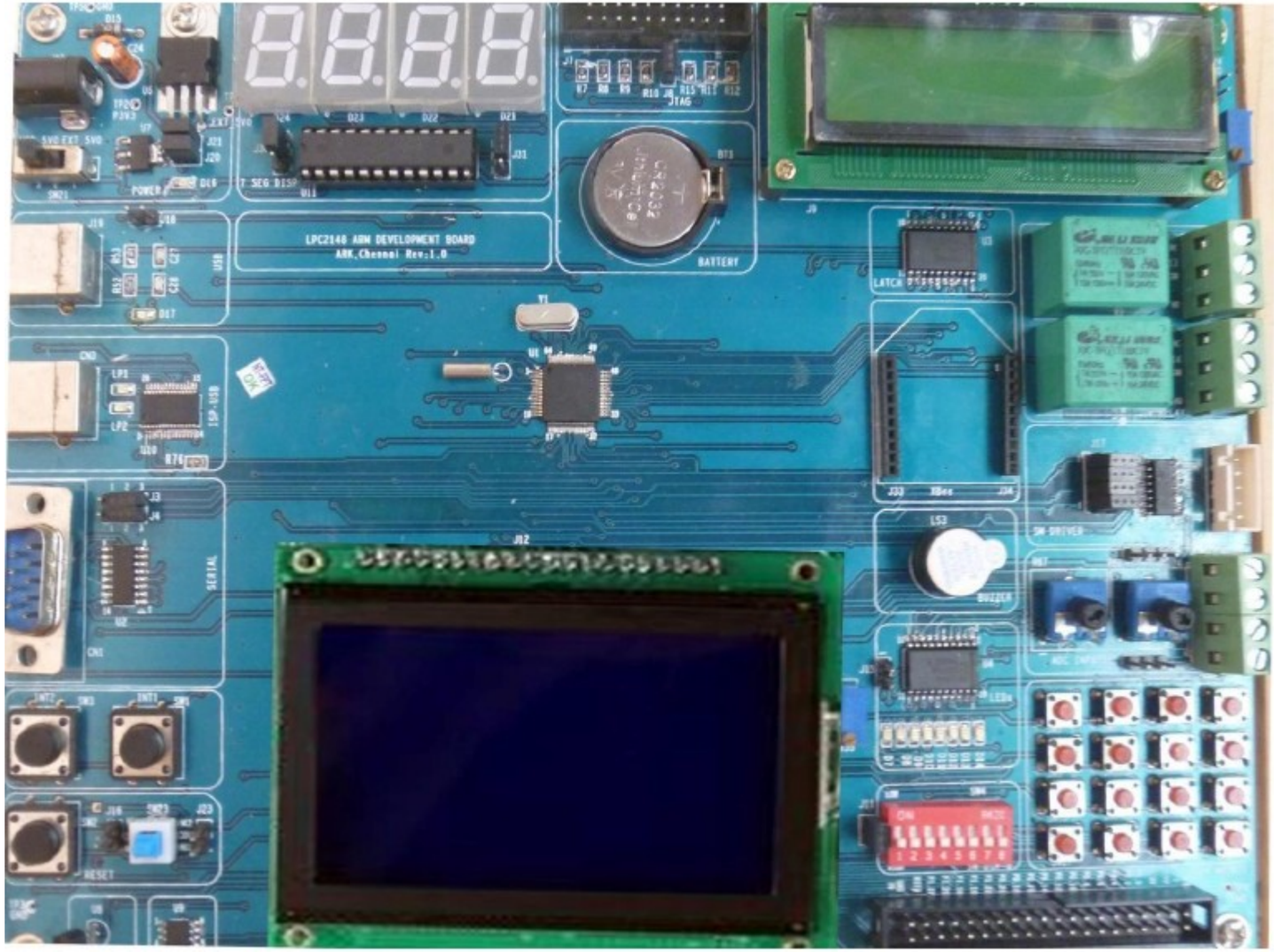
ViARM2378 ARM Development Board



In our lab ViARM2378 TFT LCD is not provided

1. NXP LPC2378 Micro controller (TQFP-144 Packaging).
2. Power supply section.
3. UART.
4. CAN Port.
5. Stepper Motor.
6. Relay.
7. USB 2.0 Device Conne
8. 10/100 Base T Ethernet Connector.
9. PS2- Keyboard connector.
10. Stereo Jack for USB Audio Device.
11. Prog/Exec Switch.
12. Joystick.
13. TFT LCD.
14. 4 x 4 Matrix Keypad.
15. 8 Way DIP switch.
16. LED.
17. SD Card Socket.
18. Analog input Trimmer.
19. 128 x 64 Pixels Graphics LCD.
20. Jtag Connector.
21. ADC, DAC and PWM Expansion slot.
22. 50Pin Expansion Header.
23. J-Trace.
24. Seven Segment Display.
25. Serial EEPROM.
26. Speaker.
27. Temperature Sensor

NXP LPC2148 Development Board

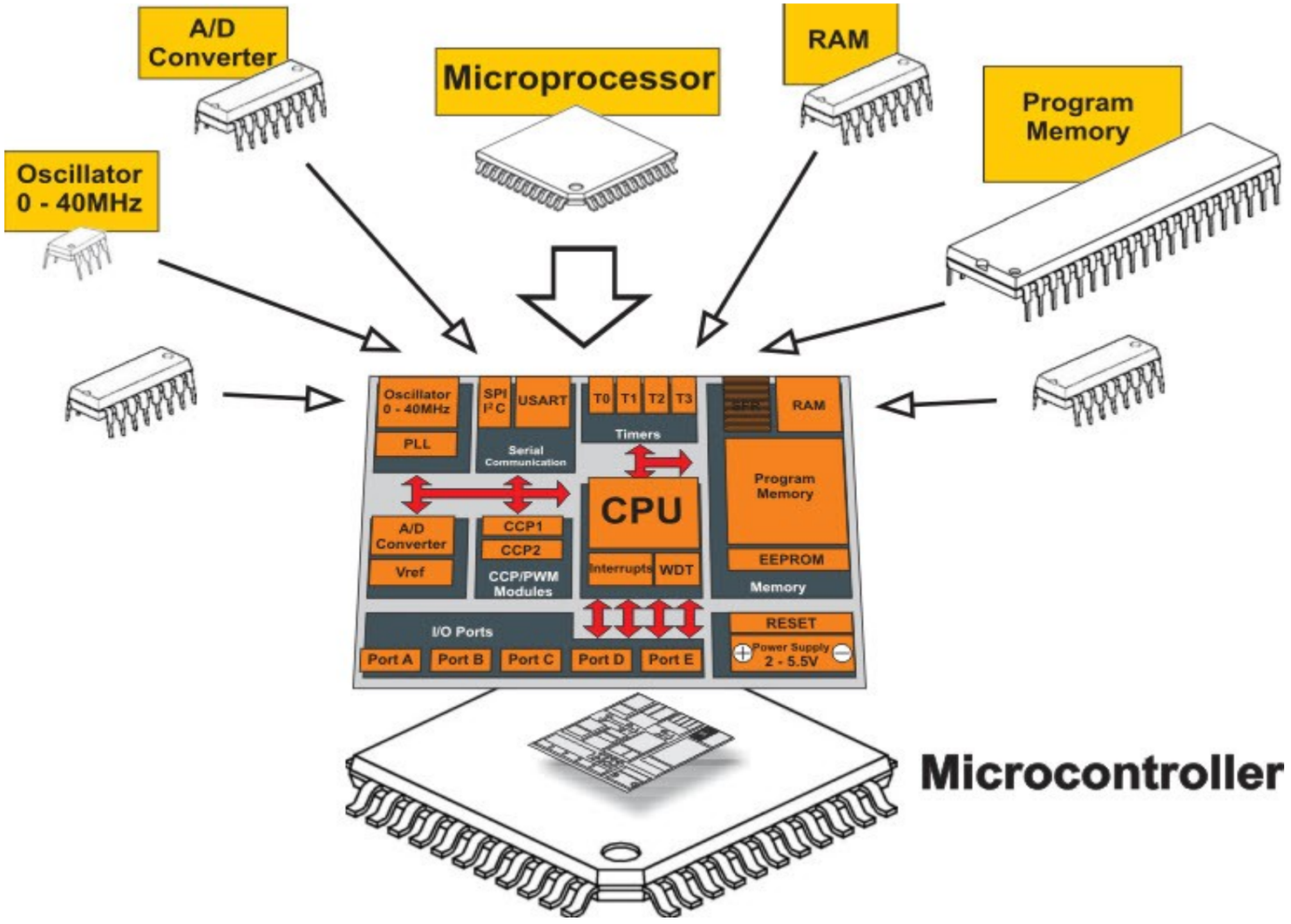


Week 2

Video Lecture Recorded on 15.8.2025

MicroController Architecture

- Microcontroller



Oscillators, RF front-end & digital Receiver, Serial communication Technologies: SPI / I2C & USART, A/D & D/A converter, I/O ports (including watch dog timer (WDT) & interrupts), Timers, Memory (RAM, EEPROM)

CCP – Capture, Compare & PWM Module

PWM – Pulse Width Modulation

Microprocessor Vs Microcontroller

- Microprocessor
  - Provides mainly the CPU
  - Without RAM, ROM and peripherals
  - General purpose computation: could be used for anything from simple to more memory / computationally intensive applications
  - Typically consumes more power than microcontroller – needs power outlets
- Microcontroller
  - A complete computer despite designed optimally for a specific task
    - This means I/O modules, RAM & ROM / EPROM are all built-in
  - Metal oxide semiconductor technology --> cheaper
  - General processing speed lower – but optimized (sometimes through hardware accelerator) for the targetted application & hence the speed is acceptable in the real-time, real-world targetted application.
  - Power consumption is lesser – often battery operated.
- Closely related terms:
  - Embedded processor, wearables, CoM, SoM, ASIC, SoC, FPGA etc

Microprocess versus Microcontroller

S No	Property	Microprocessor	Microcontroller
1	Hardware	Provides mainly CPU. This alone is useless.	A complete computer despite designed for specific application. Contains microprocessor within
2	Hardware	Without RAM, ROM and I/O [It has only CU & EU]	Includes RAM, ROM, I/O, peripherals like oscillator, PLL, A/D, D/A, timers & interrupts apart from muP
3	Purpose	General purpose computation: If interfaced could be used for anything from simple to more memory / computationally intensive applications	Specific targetted application, by & large ONLY one program in ROM
4	Power requirements	Requires larger power. Needs power outlets. Few hundred watts	Power requirements are low: Battery operated. Milli-Watts to a watt
5	Mobility	Not mobile. Can be portable. Laptops work on battery once charged. MuPs are interfaced	Typically mobile. Battery operated – battery need not be chargeable.
6	Processing speed	Typically many orders of processing speed higher as compared to that of muC	Typically the ‘embedded’ microprocessor speed is lower. But, the customized hardware in microcontroller makes the ‘custom task’s’, computational time less and comply to even real-time application requirements.
7	Power dissipated	Power dissipated culminates in the form of heat generated. Hence, heat sinks are required	Power dissipated is not perceivable. Hence heat sinks are not needed typically.
8	Programs	A program is edited, compiled, tested – design & development stage. Next program is taken for design & development. So on ... general purpose PC	Post deployment stage (Single program – permanently burnt in ROM – No change possible till the life of the product).
9	Hardware	System built (Program + data) both in RAM – main Memory [BIOS ROM & secondary storage are non-volatile]	ROM for program, RAM for data separate
10	Hardware	System built Van Neumann model	Harvard model

Architecture Versus Organization

- Microcomputer architecture
  - Attributes of computing system which are explicit during the logical execution of the program
  - Attributes visible to a programmer
  - Eg: multiplier instruction – MULT();
- Microcomputer organization
  - How the above (architectural) attribute / instruction is realized internally using hardware
  - Multiplier instruction realization
    - Either dedicated hardware (eg Booths algorithm)
    - Or using “Addition” hardware repeatedly

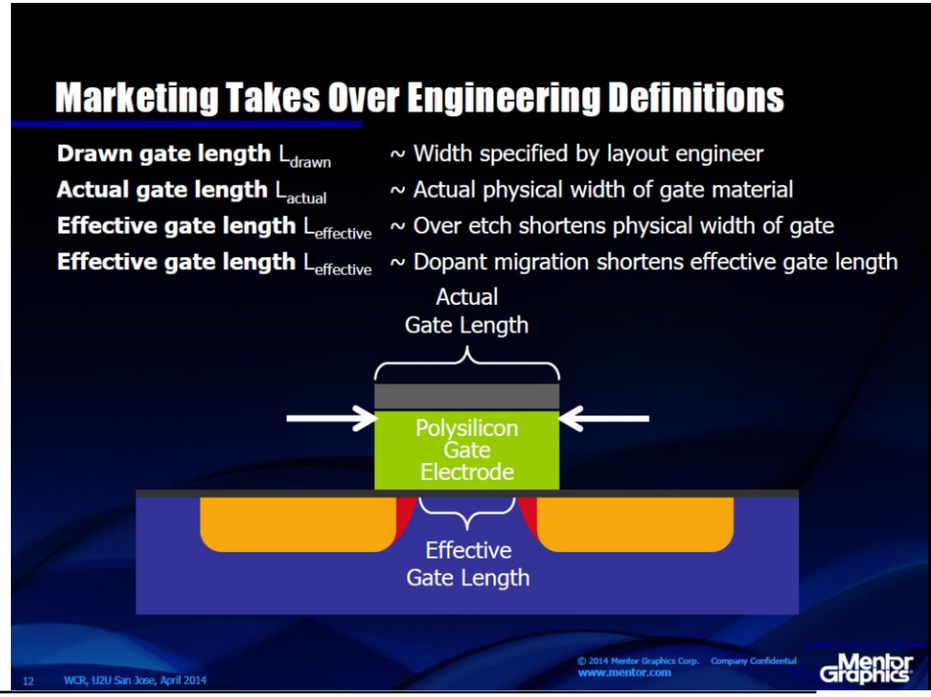
Architecture Versus Organization

- The above implementation decision based on
  - Anticipated frequency of use of the multiply instruction
  - Relative speed of two approaches (latency)
  - Cost & physical size (foot print)
- Different computer models with the same architecture (but with different organization)
  - Cost & performance (cheaper slower model, upgrade to expensive, faster models)
    - Same ISA (but, different underlying digital circuits, technology used 60 nm or 45 nm or 10 nm or 18A (1.8 nm) 2024?)
      - » 10 nm indicates the channel width in FIN-FET CMOS technology
    - Same program (compilers, IDE, softwares are same) would work for upcoming models also
  - Models changing with changing technology
    - Slower or faster models (celeron or core i9?)

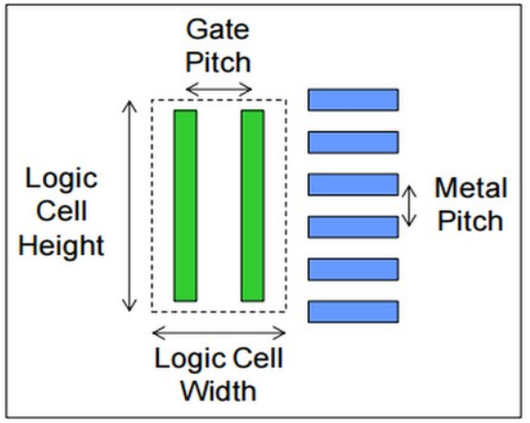
Minaturization and Moors Law

- Miniaturization
  - Pack more transistors in the same area
  - Reduce interconnect (metal) pitch and contacted gate pitch
- Moore's law
  - Moore's Law states that the number of transistors on a microchip doubles every two years
- C-MOS Technology
  - a 5 nm node is expected to have a contacted gate pitch of 51 nanometers and a tightest metal pitch of 30 nanometers
- Fabrication Process / Technology

Process	Gate pitch	Metal pitch	Year
5 nm	51 nm	30 nm	2020
3 nm	48 nm	24 nm	2022
2 nm	45 nm	20 nm	2024?

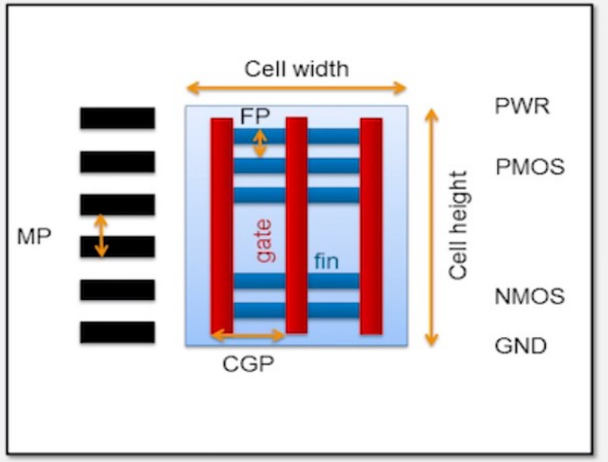


Interconnected (metal) pitch & contacted gate pitch



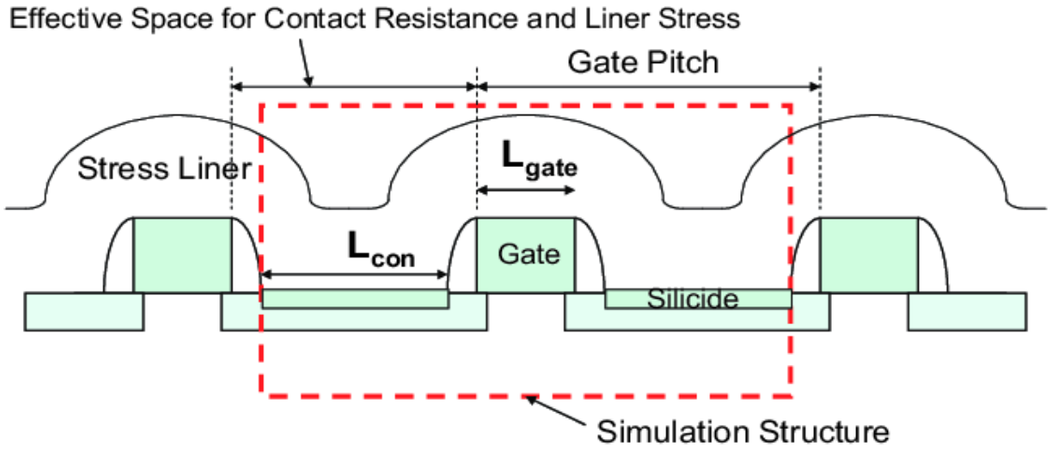
Terminology Of Scaling

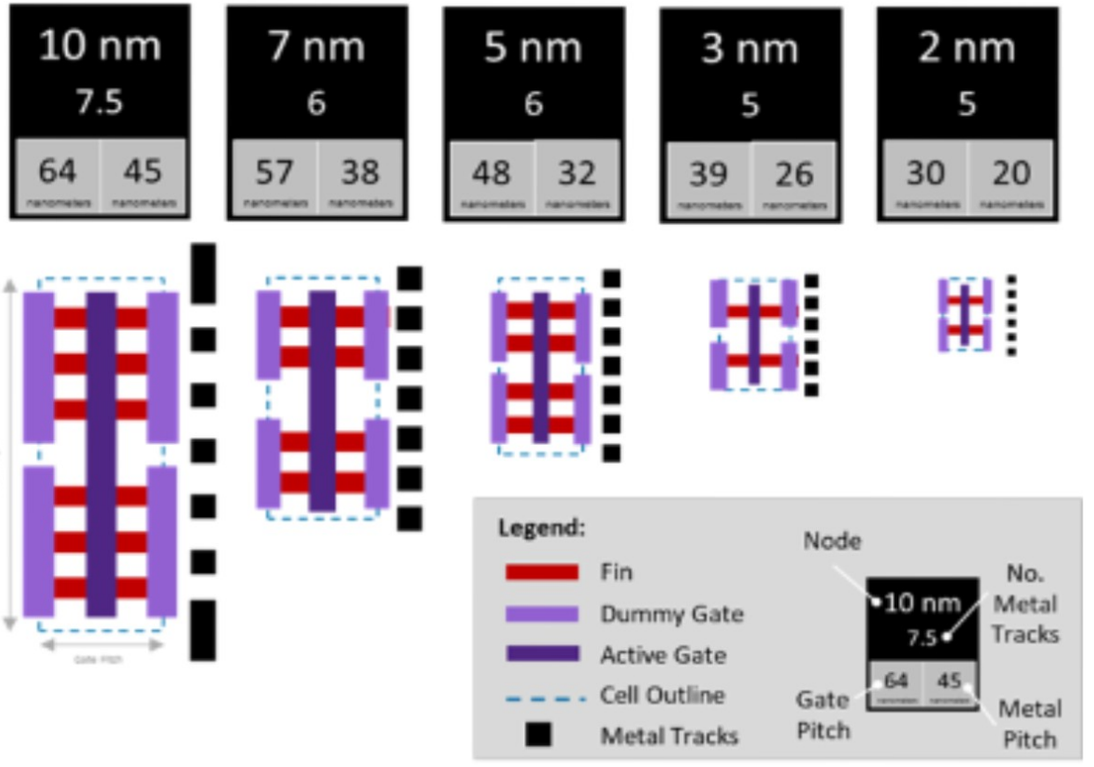
- **Logic chip** is based on std cell library
- **Std cells** defined by:
  - ***Gate pitch*** (CGP), a.k.a. CPP (Contacted Poly Pitch)
  - ***Metal pitch*** (MP)
  - ***Fin pitch*** (FP)



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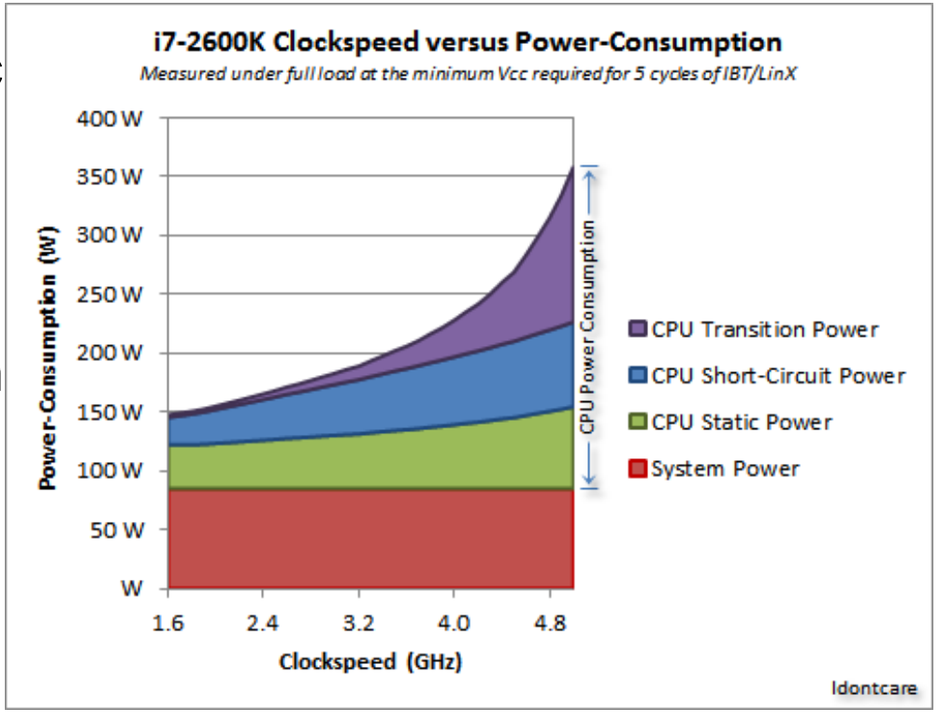
7/08/23

Frequency versus Power Dissipated

- Microprocessor Frequency
  - Clock speed
  - Basic (smallest) interval of time, any digital event can happen in a processor
  - More the frequency, more the operations per sec --> computational speed of processor
  - More number of switching per sec
- Microprocessor power
  - Inside LSI muP, digital (both combinational & sequential) circuitry have some capacitance
  - Reactive power increases when frequency increases
  - Ideally capacitor consumes zero real power
  - But, as frequency increases, (ac) current increases and in reality there is some real power consumed (small stray resistance in the branch of capacitance)
  - A rate of electrical energy supplied to muP is converted into some other form, heat, which is not used for computations → dissipated power proportional to CV^2 f

Frequency versus Power Dissipated

- Power Consumed by Capacitor
  - a practical capacitor has an imperfect dielectric
  - a small amount of power --> dielectric loss.
  - dielectric loss is almost negligible in most electrostatic capacitors.
  - In electrolytic capacitors, the dielectric loss can go from a low if 1.6 % to as high as 15 % of its reactive power for different types.
  - Stray resistance adds to real power dissipated

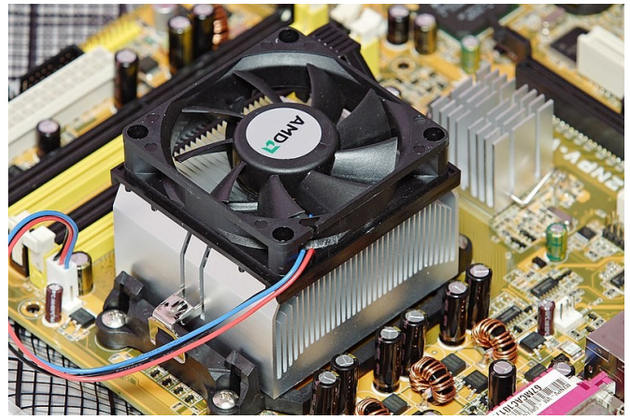


- Power dissipated --> heat generated
  - Dielectric loss + I^2 R loss (CV^2 f -) --> heat
  - Heat generated trapped due to dielectric (substrate)
  - ----> rise in temperature
  - ----> components misbehave → heat has to be thrown out
  - Cooling mechanisms

Frequency versus Power Consumed

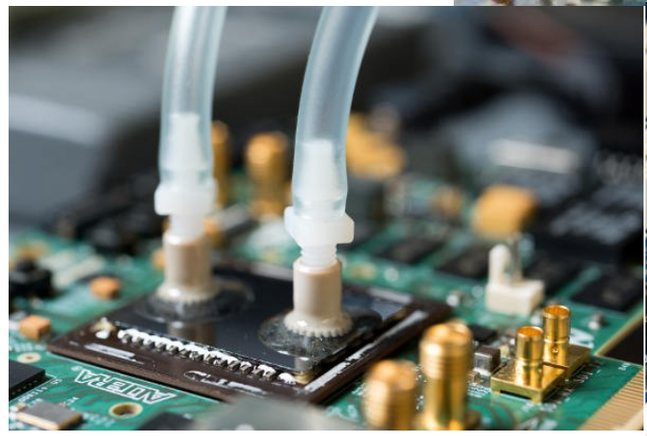
- Heat is generated
  - If not thrown away
    - temperature rises
    - --> component misbehaves
    - --> leads to glitch
      - o/p --> garbage

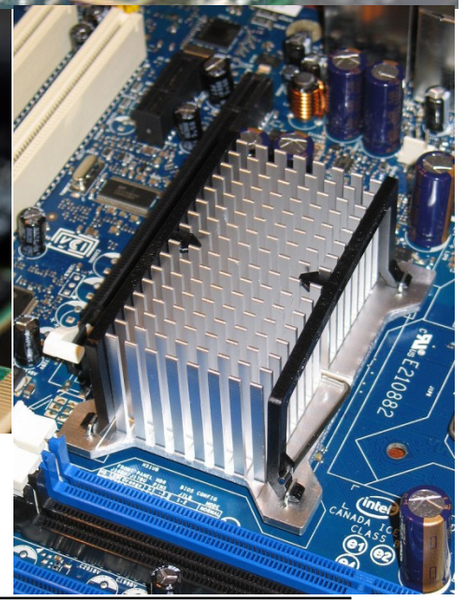






- Heat dissipating mechanisms
  - Fan
  - Al sink
  - Water cooling
  - Mineral oil immersion
    - Thermally conductive, but not electrically conductive





Classification by Model for Microcomputer

- Classification by Instruction Set Architecture (ISA)
  - Complex Instruction Set Computer (CISC) or Reduced Instruction Set Computer (RISC)
- Classification by microcomputer model
  - Van Neumann Model
  - Harvard Model

Microprocessor Vs Microcontroller

- Atmel Atmega8 is a microcontroller while ARM is a microprocessor
- ARM is a microprocessor architectural design
  - (not a physical processor, its called core) which ARM holdings inc, gives licence to others.
  - Manufacturer could fabricate either a microprocessor or microcontroller.
- Main Organizational (Hardware) Difference
  - Microcontrollers are based on Harvard model where program memory and data memory are separate.
    - Programs are stored in program memory in ROM and for throughout the life of the equipment / embedded system it is undisturbed (eg, ECU in a car), while RAM is used to store the data (eg. speed of car, temperature of running engine, coolant temperature, RPM, etc).
  - While general purpose PC (with separate microprocessor chip) are based on Von Neumann model where program and data are stored in same memory module.
    - This is developmental phase or general application wherein the program itself is continuously edited which requires RAM (volatile)
      - » may be a softcopy is stored in the secondary memory (HDD - non-volatile)